

Shubham Chaturvedi

Software Engineer

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GitHub

LinkedIn

Portfolio

SUMMARY

Software Engineer with experience in building scalable, cloud-native systems (handling **1M+ requests/day**) using Go, Node.js, PostgreSQL, AWS, and React.js, with a strong foundation in DSA, concurrency, and system design.

EXPERIENCE

Covrzy | Website

October 2024 - Present

Software Engineer

Bangalore

- Architected and optimized AWS infrastructure (EC2, S3, RDS, API Gateway, ALB, CloudWatch) for Travel APIs, integrating APIs from **India's top insurers** as downstream services for real-time insurance booking and instant policy issuance, achieving **99.9%** uptime, **<150ms** latency, and 600k monthly bookings.
- Led React web performance, SEO, and accessibility optimization, improving Lighthouse scores: **Performance 25→60, Accessibility 40→80, Best Practices 40→95, SEO 50→100** via bundle size reduction and Core Web.
- Built and load-tested a high-performance Golang and nodejs REST API, achieving **50–100ms** average latency and sustaining **10–20 TPS** (**1M+ requests/day**) under continuous traffic.

Weknow Technologies | Website

February 2023 - March 2024

Software Engineer Intern

Lucknow

- Automated the exam reevaluation module in **KNIPSS** College ERP, enabling online recheck applications and integrating CCAvenue Payment Gateway, decreasing the manual process time from **5 days to just 1 day**.
- Engineered a government survey portal, collecting data from **60,000+** users across Uttar Pradesh with multi-layer verification and automated ID card generation for secure user management.
- Developed and implemented **UPRNSS** Payroll System (HRMS) using **Agile methodologies** and **SCM** best practices, optimizing salary management for **1,000+** employees using TypeScript, MySQL, Node.js, and React.

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, Database Systems

Languages: Java, Go, TypeScript, JavaScript, SQL

Frontend: React.js, Next.js, React Native, Tailwind CSS

Backend & Frameworks: Node.js, Express.js, REST APIs, Microservices, JWT, WebSockets, Distributed Systems

Databases: PostgreSQL, MySQL, MongoDB, DynamoDB

Cloud & DevOps: AWS (EC2, RDS, S3, Lambda, API Gateway, VPC, ALB, CloudWatch), Docker, Nginx, CI/CD

System Skills: Concurrency, Caching, Rate Limiting, Load Testing, Performance Tuning, Observability

PERSONAL PROJECTS

SafeMap | Github

High-Performance Concurrent, thread safe Key-Value Store (Go)

- Engineered a thread-safe, in-memory cache using a **sharded locking strategy** (64 buckets) with **sync.RWMutex**, which reduced lock contention and maximized CPU core utilization compared to standard global locking.
- Achieved a benchmark throughput of **4,300+ JSON transactions/second** on local hardware by profiling the application and optimizing I/O-bound bottlenecks (logging and serialization).
- Designed a custom middleware pipeline for **IP-based Rate Limiting**, JWT Authentication, and **LRU-based TTL eviction** using background goroutines, ensuring O(1) access and bounded memory without external frameworks.
- Tech Stack: Go (Golang), System Design, Concurrency (Mutex/Atomic), REST API, Benchmarking tools.

CollabCode | Live | Github

Real-time collaborative code editor with WebSocket integration.

- Implemented CollabCode, a real-time code collaboration platform using **Socket.IO**, which improved team productivity by nearly 40% and reduced debugging time by 30% in collaborative environments.
- Enabled in-browser code execution, allowing users to run and test code snippets instantly without leaving the platform.
- Tech Stack: React.js v18+, Framer Motion v11+, MongoDB, Express.Js, Node.js, Socket.IO, JWT.

EDUCATION

Dr.Rammanohar Lohia Awadh University, Ayodhya

2022-24

Master of Computer Application

CGPA: 8.5

INTERESTS AND ACHIEVEMENTS

DSA: Solved 1000+ Coding problems in various Coding platforms. [LeetCode](#) | [CodeChef](#)

Technical Writing: Authored 20+ tech blogs on [Medium](#), covering web development, React optimization, system design, AWS deployment, and software engineering best practices.

Certificate: [100+ hours Java with DSA course](#) | [60+ hours Web Development Bootcamp](#) | [ChatGPT Prompt Engineering for Developers](#)

Coding Club: **Co-founded** and actively led a coding club of **250+ students**, driving the success of 7+ technical events.

Achievements: Secured **Gold Medal** in an inter-college **chess competition**.